Worksheet 6.2 — Definite Integrals and Area

Mathematics: Analysis & Approaches SL (IB) · Unit 6: Integral Calculus

The Big Ideas

A definite integral gives net signed area; true area requires splitting at every x-intercept.

- $\int_a^b f(x)\,dx = F(b) - F(a)$.
- If f changes sign in $[a, b]$, split at the intercept and take absolute values before adding.

Worked Examples

Example 1 — Basic evaluation

Evaluate $\int_1^4 (3x + 2)\,dx$.

Solution. $\left[\frac{3}{2}x^2 + 2x\right]_1^4 = (24 + 8) - (1.5 + 2)$.

Conclusion: $32 - 3.5 = 28.5$.

Example 2 — Area under a curve

Find the area under $f(x) = x^2$ from $x = 1$ to $x = 4$.

Solution. $\left[\frac{x^3}{3}\right]_1^4 = \frac{64}{3} - \frac{1}{3}$.

Conclusion: 21 square units.

Example 3 — Curve dips below the axis

Find the total area between $f(x) = x^2 - 9$ and the x-axis on $[0, 4]$.

Solution. Intercept at $x = 3$. $\int_0^3 (x^2-9)dx = 9-27 = -18$; $\int_3^4 (x^2-9)dx = (64/3-36)-(9-27) = 10/3$.

Conclusion: total $= 18 + 10/3 = 64/3$.

Example 4 — Integral properties

Given $\int_0^8 f\,dx = 25$ and $\int_0^5 f\,dx = 14$, find $\int_5^8 f\,dx$.

Solution. $25 - 14$.

Conclusion: 11.

Example 5 — Evaluating with a negative lower limit

Evaluate $\int_{-2}^3 (2x + 1)\,dx$.

Solution. $[x^2 + x]_{-2}^3 = (9 + 3) - (4 - 2)$.

Conclusion: $12 - 2 = 10$.

Example 6 — Area under a curve entirely below the axis

Find the true area between $f(x) = -x^2 - 1$ and the x-axis on $[0, 3]$ (note $f < 0$ throughout).

Solution. $\int_0^3 (-x^2 - 1)dx = \left[-\frac{x^3}{3} - x\right]_0^3 = -9 - 3 = -12$ (a negative signed area, since the curve never crosses the axis here).

Conclusion: true area $= |-12| = 12$ square units.

Example 7 — Reversing the limits of integration

Given $\displaystyle\int_2^7 f(x)\,dx = 15$, find $\displaystyle\int_7^2 f(x)\,dx$.

Solution. Swapping the limits negates the integral: $\int_7^2 f\,dx = -\int_2^7 f\,dx$.

Conclusion: -15.

Example 8 — Definite integral requiring linear substitution first

Evaluate $\displaystyle\int_1^3 (2x-1)^2\,dx$.

Solution. Antiderivative: $\frac{(2x-1)^3}{6}$. At $x = 3$: $\frac{125}{6}$. At $x = 1$: $\frac{1}{6}$.

Conclusion: $\frac{125}{6} - \frac{1}{6} = \frac{124}{6} = \frac{62}{3}$.

Example 9 — Application — accumulated rainfall

A rainfall rate is $r(t) = 4 - 0.5t$ mm/hr for $0 \le t \le 10$ hours. Find the total rainfall, and explain the meaning of the point where $r(t) = 0$.

Solution. $\int_0^{10} (4 - 0.5t)dt = [4t - 0.25t^2]_0^{10} = 40 - 25 = 15$ mm.

Conclusion: total rainfall 15 mm; $r(t) = 0$ at $t = 8$, meaning it stops raining at that point (rate becomes negative afterward, which isn't physically meaningful for rainfall, so the model likely only applies up to $t = 8$).

Your Turn — Show Your Work

For each question, show all steps clearly in the space provided.

Question 1

Evaluate $\int_0^3 (4x+1)\,dx$.

Question 2

Find the area under $f(x) = 16 - x^2$ between $x = -4$ and $x = 4$.

Question 3 — Challenge

Find the total (unsigned) area between $f(x) = x^2 - 4x$ and the x-axis on $[0, 5]$.

Question 4

Given $\int_2^{10} f\,dx = 40$ and $\int_6^{10} f\,dx = 22$, find $\int_2^6 f\,dx$.

Question 5

A tank's flow rate is $r(t) = 12 - 4t$ L/min for $0 \le t \le 5$; find the net volume change over this interval.

Question 6 — Challenge

Find the total (unsigned) area between $f(x) = x^2 - 2x - 3$ and the x-axis on $[-2, 4]$ (factor to find the intercepts first).

Question 7

Evaluate $\int_{-1}^{2} (3x^2 - 2)\,dx$.

Question 8

Find the true area between $f(x) = -x^2 - 4$ and the x-axis on $[0, 2]$.

Question 9

Given $\int_{3}^{9} f(x)\,dx = 22$, find $\int_{9}^{3} f(x)\,dx$.

Question 10

Evaluate $\int_0^2 (3x+1)^2\,dx$.

Question 11

A rainfall rate is $r(t) = 6 - t$ mm/hr for $0 \le t \le 12$; find the total rainfall and explain the meaning of where $r(t) = 0$.

Question 12 — Challenge

Find the total (unsigned) area between $f(x) = x^2 - x - 6$ and the x-axis on $[-3, 3]$ (factor first).

Question 13 — Challenge

Given $\int_1^5 f\,dx = 8$ and $\int_1^3 f\,dx = -2$, find $\int_3^5 f\,dx$, then find $\int_5^3 f\,dx$.

Question Recap & Answer Key

Question Recap

1. Evaluate $\int_0^3 (4x+1)\,dx$.
2. Find the area under $f(x) = 16 - x^2$ between $x = -4$ and $x = 4$.
3. Find the total (unsigned) area between $f(x) = x^2 - 4x$ and the x-axis on $[0, 5]$.
4. Given $\int_2^{10} f\,dx = 40$ and $\int_6^{10} f\,dx = 22$, find $\int_2^6 f\,dx$.
5. A tank's flow rate is $r(t) = 12 - 4t$ L/min for $0 \le t \le 5$; find the net volume change over this interval.
6. Find the total (unsigned) area between $f(x) = x^2 - 2x - 3$ and the x-axis on $[-2, 4]$ (factor to find the intercepts first).
7. Evaluate $\int_{-1}^2 (3x^2 - 2)\,dx$.
8. Find the true area between $f(x) = -x^2 - 4$ and the x-axis on $[0, 2]$.
9. Given $\int_3^9 f(x)\,dx = 22$, find $\int_9^3 f(x)\,dx$.
10. Evaluate $\int_0^2 (3x+1)^2\,dx$.
11. A rainfall rate is $r(t) = 6 - t$ mm/hr for $0 \le t \le 12$; find the total rainfall and explain the meaning of where $r(t) = 0$.
12. Find the total (unsigned) area between $f(x) = x^2 - x - 6$ and the x-axis on $[-3, 3]$ (factor first).
13. Given $\int_1^5 f\,dx = 8$ and $\int_1^3 f\,dx = -2$, find $\int_3^5 f\,dx$, then find $\int_5^3 f\,dx$.

✓ Answer Key

1. $[2x^2 + x]_0^3 = 18 + 3 = 21$
2. $[16x - x^3/3]_{-4}^4 = (64 - 64/3) - (-64 + 64/3) = \frac{256}{3}$
3. intercepts $0, 4$; $\int_0^4 (x^2 - 4x)dx = [\frac{x^3}{3} - 2x^2]_0^4 = \frac{64}{3} - 32 = -\frac{32}{3}$; $\int_4^5 (x^2 - 4x)dx = (\frac{125}{3} - 50) - (\frac{64}{3} - 32) = \frac{61}{3} - 18 = \frac{7}{3}$; total $= \frac{32}{3} + \frac{7}{3} = 13$
4. $40 - 22 = 18$
5. $[12t - 2t^2]_0^5 = 60 - 50 = 10$ L
6. intercepts $x = -1, 3$; $\int_{-2}^{-1} (x^2 - 2x - 3)dx = (-\frac{1}{3} - 1 + 3) - (-\frac{8}{3} - 4 + 6) = \frac{5}{3} - (-\frac{2}{3}) = \frac{7}{3}$; $\int_{-1}^3 (\cdot)dx = (9 - 9 - 9) - (-\frac{1}{3} - 1 + 3) = -9 - \frac{5}{3} = -\frac{32}{3}$; $\int_3^4 (\cdot)dx = (\frac{64}{3} - 16 - 12) - (9 - 9 - 9) = -\frac{8}{3} + 9 = \frac{19}{3}$; total $= \frac{7}{3} + \frac{32}{3} + \frac{19}{3} = \frac{58}{3}$
7. $[x^3 - 2x]_{-1}^2 = (8 - 4) - (-1 + 2) = 4 - 1 = 3$
8. $\int_0^2 (-x^2 - 4)dx = [-\frac{x^3}{3} - 4x]_0^2 = -\frac{8}{3} - 8 = -\frac{32}{3}$; true area $= \frac{32}{3}$
9. -22
10. antiderivative $\dfrac{(3x+1)^3}{9}$; at $x = 2$: $\dfrac{343}{9}$; at $x = 0$: $\dfrac{1}{9}$; result $= \dfrac{342}{9} = 38$
11. $\int_0^{12} (6 - t)dt = [6t - t^2/2]_0^{12} = 72 - 72 = 0$; net rainfall is 0, since $r(t) < 0$ after $t = 6$ isn't physical — the model only really applies up to $t = 6$, giving true rainfall $= \int_0^6 (6 - t)dt = 36 - 18 = 18$ mm
12. intercepts $x = -2, 3$; $\int_{-3}^{-2} (x^2 - x - 6)dx = (-9 - 4.5 - 18) - (-9 - 2 - 18\ldots)$ carefully: $F(x) = \frac{x^3}{3} - \frac{x^2}{2} - 6x$; $F(-3) = -9 - 4.5 + 18 = 4.5$, $F(-2) = -\frac{8}{3} - 2 + 12 = 7.33$, $F(3) = 9 - 4.5 - 18 = -13.5$; $\int_{-3}^{-2} = 7.33 - 4.5 = 2.83$; $\int_{-2}^3 = -13.5 - 7.33 = -20.83$; $\int_3^3$ n/a; total $= 2.83 + 20.83 \approx 23.67$
13. $\int_3^5 f = 8 - (-2) = 10$; $\int_5^3 f = -10$